

$$1. \quad \frac{2xy}{x^2+1} - 2x - (2 - \ln(x^2+1))y' = 0$$

$$\left(\frac{2xy}{x^2+1} - 2x \right) dx - (2 - \ln(x^2+1)) dy = 0$$

$$\left(\frac{2xy}{x^2+1} - 2x \right)'_y = \frac{2x}{x^2+1} = \left(-(2 - \ln(x^2+1)) \right)'_y$$

$$u(x,y) = \int \frac{2xy}{x^2+1} - 2x dx = y \ln(x^2+1) - x^2 + g(y)$$

$$\ln(x^2+1) - 2 = u'_y = \ln(x^2+1) + g'(y)$$

$$g'(y) = -2 \quad g(y) = -2y + C$$

$$u(x,y) = y \ln(x^2+1) - x^2 - 2y + C$$

$$y \ln(x^2+1) - x^2 - 2y = C$$

$$y = \frac{C + x^2}{\ln(x^2+1) - 2}$$

$$0 = y(5) = \frac{C + 25}{\ln(26) - 2}$$

$$C = -25$$

$$y = \frac{-25 + x^2}{\ln(x^2+1) - 2}$$

$$\ln(x^2+1) \neq 2$$

$$x^2+1 \neq e^2$$

$$x \neq \pm \sqrt{e^2 - 1}$$

$$\sqrt{e^2 - 1} < x$$

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$$y''' - 12y'' + 48y' - 64y = 2e^{4x} \quad .2$$

$$y = e^{4x}$$

$$r^3 - 12r^2 + 48r - 64 = \quad .1$$

$$= (r-4)(r^2 + 8r + 16) = (r-4)^3$$

$$y_H = C_1 e^{4x} + C_2 x e^{4x} + C_3 x^2 e^{4x}$$

$$y_p = a x^3 e^{4x} \quad .N$$

$$6a e^{4x} = 2e^{4x}$$

$$a = 1/3$$

$$y_p = \frac{1}{3} x^3 e^{4x}$$

$$y(x) = C_1 e^{4x} + C_2 x e^{4x} + C_3 x^2 e^{4x} + \frac{1}{3} x^3 e^{4x}$$

$$C_1 = 1 \quad C_2 = -4 \quad C_3 = 8$$

$$y(x) = e^{4x} - 4x e^{4x} + 8x^2 e^{4x} + \frac{1}{3} x^3 e^{4x}$$

$$y'' + x^2 y' - 4xy = 0 \quad .3$$

$$\sum_{n=0}^{\infty} (n+1)(n+2) a_{n+2} x^n + \sum_{n=0}^{\infty} n a_n x^{n+1} + \sum_{n=0}^{\infty} -4 a_n x^{n+1} = 0$$

$$\sum_{n=0}^{\infty} (n+1)(n+2) a_{n+2} x^n + \sum_{n=1}^{\infty} (n-1) a_{n-1} x^n + \sum_{n=1}^{\infty} -4 a_{n-1} x^n = 0$$

$$2a_2 + \sum_{n=1}^{\infty} \left((n+1)(n+2) a_{n+2} + (n-1) a_{n-1} - 4 a_{n-1} \right) x^n = 0$$

$$a_2 = 0$$

$$a_{n+2} = - \frac{n-5}{(n+1)(n+2)} a_{n-1} \quad n \geq 1$$

$$a_2 = 0 \Rightarrow a_{2+3k} = 0 \quad k \geq 0 \quad .2$$

$$y(0) = a_0 = 0 \Rightarrow a_{3k} = 0 \quad k \geq 0$$

$$y'(0) = a_1 \Rightarrow a_4 = - \frac{-3}{3 \cdot 4} \cdot a_1 = + \frac{1}{4} a_1$$

$$a_2 = 0 \Rightarrow a_{1+3k} = 0 \quad k \geq 2$$

$$y(x) = y'(0)x + \frac{1}{4} y'(0)x^4 = y'(0) \left(x + \frac{1}{4} x^4 \right)$$

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$$y'(0) \neq 0, \quad y(0) = 0$$

$$\mathcal{L}\{u_c(t) e^{-bt}\}(s) = \frac{e^{-c(s+b)}}{s+b} \quad .10 .4$$

$$y'' + 3y' + 2y = u_1 \quad .2$$

$$y(0) = 1 \quad y'(0) = 0$$

$$s^2 \mathcal{L}\{y\}(s) - sy(0) - y'(0) + 3(s \mathcal{L}\{y\}(s) - y(0)) + 2 \mathcal{L}\{y\}(s) = \frac{e^{-cs}}{s}$$

$$(s^2 + 3s + 2) \mathcal{L}\{y\}(s) - s - 3 = \frac{e^{-cs}}{s}$$

$$\mathcal{L}\{y\}(s) = \frac{e^{-cs}}{s(s+1)(s+2)} + \frac{s+3}{(s+1)(s+2)}$$

$$\mathcal{L}\{y\}(s) = e^{-cs} \left[\frac{1/2}{s} - \frac{1}{s+1} + \frac{1/2}{s+2} \right] + \frac{2}{s+1} - \frac{1}{s+2}$$

$$y(t) = u_1(t) \left[\frac{1}{2} - e^{-(t-1)} + \frac{1}{2} e^{-2(t-1)} \right] + 2e^{-t} - e^{-2t}$$

$$X' = \begin{pmatrix} -5 & 1 \\ 4 & -2 \end{pmatrix} X + e^{2t} \begin{pmatrix} 6 \\ -1 \end{pmatrix} \quad .5$$

$$p(r) = (r+6)(r+1) \quad .10$$

$$-6: \begin{pmatrix} 1 & 1 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$-1: \begin{pmatrix} -4 & 1 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$$X_H = C_1 e^{-6t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + C_2 e^{-t} \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$$X_p = C_1(t) e^{-6t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + C_2(t) e^{-t} \begin{pmatrix} 1 \\ 4 \end{pmatrix} \quad .2$$

$$C_1'(t) e^{-6t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + C_2'(t) e^{-t} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = e^{2t} \begin{pmatrix} 6 \\ -1 \end{pmatrix}$$

$$C_2'(t) e^{3t} \rightarrow C_2(t) = \frac{1}{3} e^{3t}$$

$$C_1'(t) = 5 e^{8t} \rightarrow C_1(t) = \frac{5}{8} e^{8t}$$

$$X_p(t) = e^{2t} \begin{pmatrix} 23/24 \\ 17/24 \end{pmatrix}$$

$$X = C_1 e^{-6t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + C_2 e^{-t} \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \frac{1}{24} e^{2t} \begin{pmatrix} 23 \\ 17 \end{pmatrix}$$

$$X(0) = \frac{5}{8} \quad \frac{35}{24}$$

$$X = -\frac{1}{4} e^{-6t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \frac{1}{8} e^{-t} \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \frac{e^{2t}}{24} \begin{pmatrix} 23 \\ 17 \end{pmatrix}$$